

Bypassing Conventional eKYC: How Far Can We Go using Deepfake?

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Abstract

This study examines the growing threat that deepfake technology poses to cybersecurity, particularly with regard to electronic Know Your Customer (eKYC) systems. Deepfakes, which are produced using sophisticated AI known as Generative Adversarial Networks (GANs), may produce lifelike fake faces that make it challenging to accurately confirm someone's identity. In order to fool eKYC systems that rely on liveness checks, we employed the Buffalo_1 model to identify faces and the inswapper_128 model to produce deepfake videos from a single image. We then put the videos to the test on a number of websites that are meant to identify fakes or verify whether a person is in fact real. Our research uncovers significant weaknesses in the way eKYC security currently operates, demonstrating the pressing need for improved methods to identify these deepfakes. We also analyze the models used in the deepfake production process, as well as the performance of several detection tools.

Keywords

Deepfake, eKYC, Liveliness Detection, Cybersecurity

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1 Introduction

The emergence of Deepfake technology is one of the most concerning developments as artificial intelligence continues to advance at a remarkable pace[19]. Deepfakes employ advanced deep learning techniques to produce incredibly lifelike fake audio and video that can give the impression that someone said or did something they never did. The ease of access to this technology is what makes it even more frightening. These days, anyone can create convincing deepfake content of people, whether they are public personalities or private residents, with only a few clicks and user-friendly software or apps. No technical expertise is needed. The attack surface has

grown beyond earlier borders as more people post their images on social media.[21]

This ease in creating deepfakes has major consequences. Impersonation attacks can be extremely dangerous when someone's voice, physical features, or behavior can be convincingly replicated. Deepfake videos can be misused by malicious actors to spread misleading information, ruin people's reputations, or even carry out different kinds of fraud, like identity theft or financial schemes. For instance, deploying a deepfake to pose as a government official or business executive could deceive workers or citizens into making poor choices. The validity of many digital communication channels is called into question by these impersonation attacks, that weaken our trust in what we see and hear.

Identity verification in the digital sphere, particularly through procedures known as electronic Know Your Customer, is one area where this problem is particularly severe. Businesses can expedite access to banking, financial services, and other necessary transactions by using eKYC to remotely verify a customer's identity. By eliminating the need for in-person verification, eKYC improves security and convenience. Consumers may now authenticate themselves online using their ID documents and facial recognition technology, eliminating the need for them to physically visit a branch.

However, the current strategy faces a serious threat from the development of deepfake technology[1]. eKYC systems that rely on video or biometric verification may be tricked if someone can produce an almost perfect video that poses as a real customer. The fundamental confidence and dependability of digital identity verification are at danger. Unauthorized account openings, fraudulent transactions, and a general lack of trust in digital services could result from these systems being infiltrated.

1.1 Research Questions

RQ1 Can eKYC systems be effectively bypassed using advanced deepfake technologies?

RQ2 What are the limitations of existing liveness and deepfake detection mechanisms in identifying synthetic identities?

1.2 Core Contributions

- Implementation of the inswapper_128 model combined with the buffalo_1 model for high-accuracy deepfake generation.
- First-of-its-kind assessment of deepfake technology's ability to exploit vulnerabilities in a deployed eKYC system in Bangladesh.

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- Evaluation of generated deepfakes against publicly available liveness and deepfake detection platforms, highlighting existing gaps in detection.

2 Motivation

Deepfake technologies have the ability to take advantage of weaknesses in any digital identity detection system since they are driven by sophisticated machine learning models. The need of resolving these issues has been further brought to light by recent instances of fraud involving synthetic identities.

The way consumers confirm their identities remotely has evolved dramatically due to the growing use of eKYC systems for identity verification in sectors including banking, telecommunications, and e-commerce. Although these systems increase convenience and reduce personal contacts, they are intrinsically susceptible to new technologies such as deepfake videos. The goal of this work is to close the gap between the lack of reliable detection techniques in eKYC applications and technological developments in deepfake generation.

- **Strengthening eKYC Systems:** By identifying potential vulnerabilities, this work helps improve the design of secure identity verification processes.
- **Advancing Deepfake Detection:** The findings contribute to developing more effective algorithms for identifying synthetic media.
- **Cybersecurity Enhancements:** Insights from this study can be applied to broader domains, including financial fraud prevention and secure remote authentication.
- **Ethical AI Usage:** Highlighting the misuse of deepfake technology can encourage ethical considerations in AI development and deployment.

3 Related Work

Biometric systems, particularly facial recognition, have advanced significantly. But they have also revealed serious weaknesses to presentation attacks, such as advanced deepfake manipulations. In recent literature we see that there is an ongoing arms race between increasingly realistic deepfake generation techniques and the urgent need for robust detection mechanisms.

Advanced techniques such as *GHOST* [12] incorporate complex face-swapping pipelines that integrate identity encoding, adaptive blending, and eye-focused loss functions to give seamless facial manipulations. Both generation and detection depend on precise facial alignment, and *RetinaFace* [10] offers a unified framework for 3D vertex regression, landmark localization, and face identification. Furthermore, *Facial Geometric Detail Recovery* [25] achieves previously unheard-of levels of generation realism by simulating fine facial textures and detailed 3D geometry through physically based rendering and implicit representations. This has led to a diversification of detection efforts. Baseline surveys [20, 23] comprehensively classify attack types, including 2D, 3D, and deepfake-based presentation attacks, emphasizing the pressing need for a variety of high-quality datasets. When faced with high-fidelity deepfakes, the *eKYC-DF* dataset [11] includes over 228,000 real and synthetic facial videos which shows significant performance disparities in existing systems. Many supervised detection frameworks are supported by

large-scale datasets like *FaceForensics++*, Rössler et al., [26], which offer over 1.8 million altered photos spanning several cutting-edge forgery techniques and compression levels. This work demonstrates that including domain-specific knowledge, such as strong face tracking, can greatly increase detection accuracy beyond that of human observers, with notable performance from architectures like XceptionNet. Novel paradigms for detection are still being developed. By concentrating on dependable visual cues, such as eyes and facial hair, Chakraborty et al.'s [6] *TruthLens* reframes deepfake detection as a training-free visual question answering pipeline using large vision-language and language models, providing interpretable and adaptable classification that outperforms conventional CNN methods on difficult datasets. While blockchain-based verification frameworks Hasan and Salah [14]; Chan et al. [7] offer promising decentralized provenance tracing to authenticate video origins, ensemble architectures, exemplified by *DeepfakeStack* Rana and Sung [24], achieve near-perfect accuracy by combining multiple deep learning models.

Temporal and physiological factors provide useful indications for detection. By using a long-term recurrent neural network to monitor decreased eye blinking frequency—a minor but reliable sign of deepfake videos—Li et al. [18] take advantage of physiological irregularities. In order to solve the frame-level and sequential irregularities seen in video deepfakes, Saika et al. [27] and Guera et al. [13] propose coupled CNN-LSTM networks to capture spatial features and temporal dependencies. To capture temporal distortions and inconsistencies across frames, Tariq et al. [28] mix residual networks and convolutional LSTM. By examining photo response non-uniformity (PRNU), or distinct camera sensor noise patterns, Koopman et al. [16] provide a new statistical method that can be used in conjunction with learning-based techniques to statistically correlate and validate video authenticity.

Convolutional Neural Networks (CNNs) continue to be the most popular method for detecting spatial anomalies [22] because of its ability to detect pixel inconsistencies, blending artifacts, and artificial textures. It has been shown that GAN-based face swapping can readily produce high-quality Deepfake videos that can fool cutting-edge face recognition systems like VGG and FaceNet, underscoring the critical need for reliable detection mechanisms, especially since training and blending parameters have a substantial impact on video realism. [citekorshunov2018deepfakes](https://arxiv.org/abs/2018deepfakes)

However, inadequate temporal sensitivity and video compression frequently limit their performance. In order to improve overall robustness, hybrid models that combine recurrent neural networks (RNNs), long short-term memory (LSTM) units, and attention mechanisms have been created. These models take use of temporal correlations between video frames. The goal of more modern architectures that make use of explainable AI and transformers is to improve interpretability and long-range dependency modeling, which are essential for practical implementation.

When taken as a whole, these studies highlight a multifaceted strategy that combines advanced deep learning architectures, statistical forensics, physiological and temporal feature exploitation, dataset expansion, and interpretability. In an increasingly manipulated digital environment, this all-encompassing viewpoint is essential to outperforming advanced forgery techniques and protecting biometric systems.

4 Methodology

This section describes the architecture, data gathering, and procedure utilized to create extremely lifelike deepfake videos utilizing cutting-edge machine learning models. The three main parts of our approach are the deepfake generating process, data collecting, and model architecture.

4.1 Model Architecture

We used a mix of the buffalo_l recognition model and the inswapper_128 model to create hyper-realistic deepfake videos. When combined, these models guaranteed excellent face swapping while preserving the switched faces' identities throughout the whole video.

- **Deepfake Generation Model (inswapper_128)** A cutting-edge face-swapping framework, the inswapper_128 model expands on the InsightFace [15] paradigm of high-accuracy face recognition and embedding creation. The conditional autoencoder architecture serves as the foundation for this high-fidelity face-swapping model. Its fundamental architecture makes use of an identity encoder and a face generator, with the decoder receiving identity embeddings that are taken from a pre-trained recognition network (like buffalo_l). This guarantees that the target image's natural appearance is preserved while the synthesized face maintains its original identity. Large-scale, diversified face datasets that offer high-resolution photos with a wide range of age, position, expression, and lighting are used to train the model. Furthermore, inswapper_128 can effectively generalize to faces in the wild and retain resilience against occlusions, subtle facial differences, and compression artifacts thanks to pretraining on these datasets. Inswapper_128 is especially useful for applications in video and image-based identity synthesis because it produces realistic face swaps that are both visually cohesive and identity-consistent by fusing identity conditioning with high-resolution face generation.
 - **Encoder:** Convolutional and residual blocks make up the encoder, which gradually downsamples the source face. This produces a compact latent representation by extracting hierarchical facial features while lowering spatial resolution.
 - **Feature Fusion:** The identification embedding from buffalo_l is integrated with the encoded source face features at the latent level. By keeping pose, lighting, and expression from the source, this fusion enables the network to condition the generation process on the target identity.
 - **Decoder:** The decoder uses residual blocks and upsampling to replicate the encoder. Using skip connections from the encoder, it reconstructs the face image from the fused representation while maintaining fine-grained spatial details. Sharper and more reliable results are guaranteed by this U-Net-inspired design.
 - **Identity Consistency:** By comparing generated faces with the goal identity embedding, the model uses an identity loss. This guarantees that the swapped face stays true

to the target identity by enforcing robust identity maintenance.

- **Output Layer:** For realistic face replacement, a 128x128 RGB image of the switched face is produced by a final convolutional layer. This image can then be blended into video frames.
- **Recognition Model (buffalo_l)** Buffalo_l model is an advanced model that detects and analyzes faces in video frames. It uses a combination of CNN and attention mechanisms to accurately locate facial landmarks and optimize face alignment.
 - **Face Detection:** The buffalo_l model utilizes a multi-task cascaded CNN (MTCNN) for face detection. This approach ensures accurate identification and alignment of faces. Even under challenging conditions such as varying orientations or poor lighting, this model performs well.
 - **Face Analysis:** Attention module can accurately identify critical facial regions, such as eyes, nose, and mouth. This module enhances the model's ability to extract meaningful features, which are crucial for high-quality face swapping and alignment.

4.2 Data Collection

We sourced participant data and templates to produce realistic deepfakes. The fixed template for all deepfake-generating studies was an actual demo video from a liveness-detecting website. We obtained high-quality frontal facial photos of people facing the camera with the permission of five volunteers. We were unable to enlist more people because of privacy issues and the delicate nature of deepfake technology. We used publically accessible facial datasets to supplement the data. Nevertheless, the success of getting around such detectors is limited because many of these datasets are already being used to train detection algorithms.

4.3 Deepfake Generation Process

We just feed each volunteer's image into the pipeline together with the template video to create deepfakes. In our scenario, we fixed the template video for every individual. The entire deepfake generation pipeline is shown in Figure 2.

- **Split:** The OpenCV package was used to divide the input template video into separate frames. To ensure smooth synthesis and accurate face switching, each frame was processed separately.
- **Face Detection:** Faces were identified and aligned within every video frame using the buffalo_l model. To enable precise face switching and alignment, facial markers including the lips, nose, and eyes were recognized.
- **Face Swapping:** The swapped faces are produced by the inswapper_128 model, which incorporates the identity layer to maintain identity characteristics. Each person's identification was maintained, and their actions were modified according to their facial landmarks.
- **Video Synthesis:** OpenCV was used to compile the altered frames into a continuous video. In order to ensure flawless transitions between successive frames, the switched faces

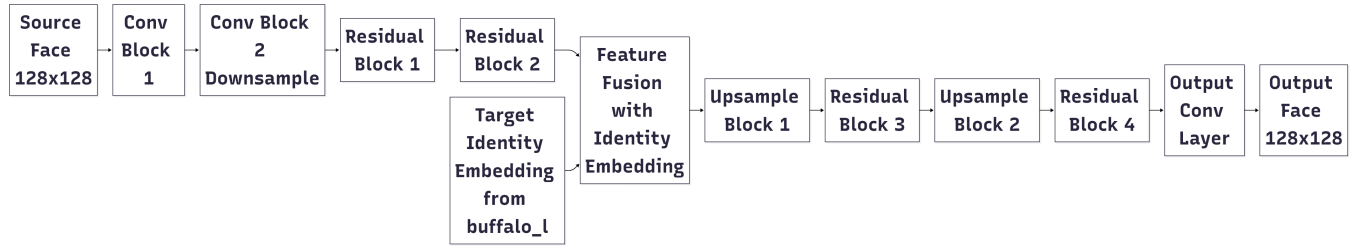


Figure 1: Simplified architecture of the inswapper_128 model, highlighting the encoder, feature fusion with identity embedding, and decoder.

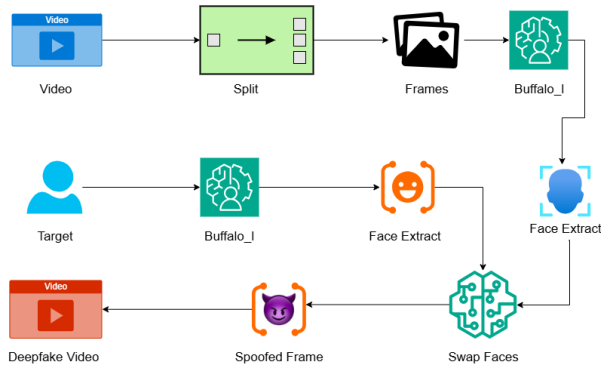


Figure 2: Deepfake generator pipeline

had to be blended into the original frames. A high-resolution, incredibly lifelike deepfake video was the end result.

4.4 Challenges and Adaptations

During the process, we encountered several challenges. We addressed them as:

- The identity layer proved essential in maintaining the integrity of participant identities in the deepfakes, ensuring high authenticity.
- Strict ethical guidelines were followed during data collection to address privacy concerns, and only consenting participants were included in the study.

5 Evaluation

We evaluated our deepfake videos on five online deepfake detectors. We found most of our videos could bypass the the detectors, questioning their reliability.

5.1 Overall Results

We tested our produced deepfakes on deepfake and liveness detectors of Cell-oscope, Bio-ID, Arya.ai, Attestive.com, Deepfake-detect.com, and Deepware.com. The findings are summarized in Table 1. We can see that many of the produced videos are able to evade specific detection procedures. Only Attestive.com's detector was able to distinguish between real and fake with a small margin.

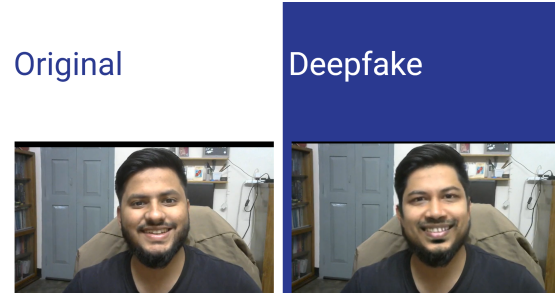


Figure 3: Original vs Fake

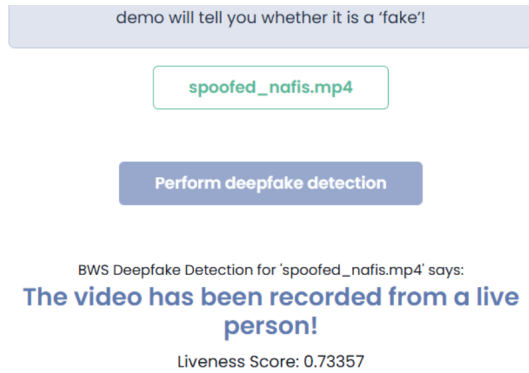
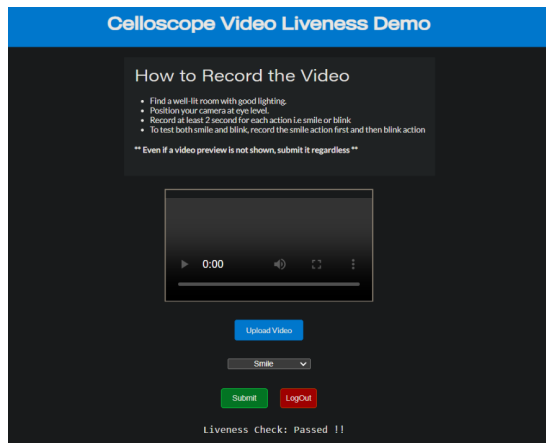
5.2 Snapshots:

These are some snapshots of our generated deepfakes and the outcomes from different online detectors. These examples provide a visual comparison of the original videos and their corresponding deepfake versions, highlighting the effectiveness of our generation pipeline in bypassing detection mechanisms.

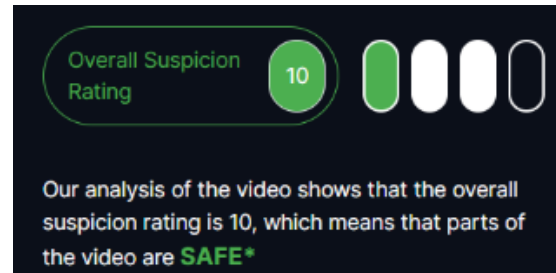
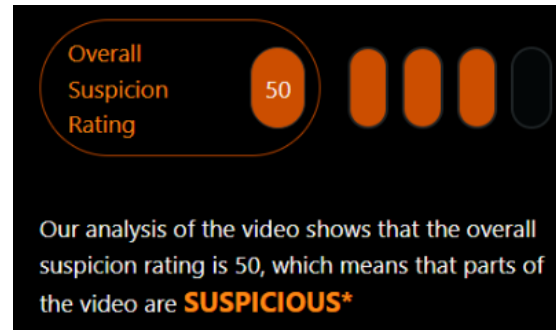
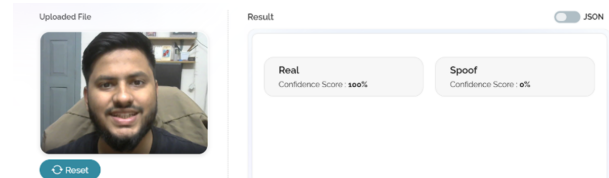
- **Original and deepfake comparison:** A side-by-side comparison 3 of original videos and their deepfake counterparts illustrates the high realism achieved by the generated deepfakes, making them difficult to distinguish from genuine videos.
- **Bio-ID deepfake detection score:** Bio-ID [4] marks a video deepfake if its score is below 0.5. With our generated deepfakes 4, the score is always higher than 0.55, indicating that they passed as authentic in all cases. 4
- **Bypassing Celloscope liveness detection:** Celloscope [5] provides liveness detection on recorded video. Our deepfakes successfully bypassed 5 the liveness detection system of Celloscope in multiple instances, showcasing the vulnerabilities of liveness verification mechanisms. 5
- **Attestive.ai deepfake detection score:** Attestive.ai [3] analyses the recorded video and scores it for safe 6, suspicious, and fake. The Attestive.ai tool demonstrated varying levels of detection accuracy. All of the deepfakes were placed in the suspicious category 7 with a score of 50 to 53, with higher-resolution videos often evading detection while lower-resolution versions were more frequently identified as fake.
- **Arya.ai liveness detection score original:** Arya.ai [2] detects if an image is forged or not. This site only allows

Table 1: Performance of Deepfake Detection Tools on Original and Spoofed Videos

Tool	Original Video	Spoofed_1	Spoofed_2	Spoofed_3	Spoofed_4	Spoofed_5
Cell-oscope	Passed	Passed	Passed	Passed	Passed	Passed
Bio_ID	Passed (Liveness Score: 0.96242)	Passed (Liveness Score: 0.6807)	Passed (Liveness Score: 0.73357)	Passed (Liveness Score: 0.57635)	Passed (Liveness Score: 0.6183)	Passed (Liveness Score: 0.68765)
Arya.ai	Score (real: 100, spoof: 0)	Score (real: 100, spoof: 0)	Score (real: 100, spoof: 0)	Score (real: 91, spoof: 9)	Score (real: 88, spoof: 12)	Score (real: 100, spoof: 0)
Attestive.com	Safe (score: 10)	Suspicious (score: 53)	Suspicious (score: 50)	Suspicious (score: 52)	Suspicious (score: 52)	Suspicious (score: 51)
Deepfake-detect.com	Prediction score: 100%	Prediction score: 98.85%	Prediction score: 99.76%	Prediction score: 99.85%	Prediction score: 99.85%	Prediction score: 100%
Deepware.com	Avatarify (Suspicious: 68%), Deepware (No deepfake: 0%)	Avatarify (No deepfake: 34%), Deepware (Deepfake: 94%)	Avatarify (No deepfake: 44%), Deepware (Deepfake: 97%)	Avatarify (Suspicious: 77%), Deepware (Suspicious: 68%)	Avatarify (Suspicious: 51%), Deepware (Suspicious: 58%)	Avatarify (No deepfake: 47%), Deepware (Suspicious: 51%)

**Figure 4: Bio-id spoofed result****Figure 5: Celloscope spoofed result**

screenshots/images, not video. We have tested with different screenshots of our deepfakes 9, and all of them achieved an equal liveness score to the original 8. 8

**Figure 6: Original video Deepfake score from attestive****Figure 7: Fake video Deepfake score from attestive****Figure 8: Original video liveness score from arya**

- **Deepfake-detect.com Prediction:** Same as Arya.ai, in Deepfake-detect.com [8] we tested our deepfakes 11 with original video 11, taking snapshots, and achieved equivalent results.

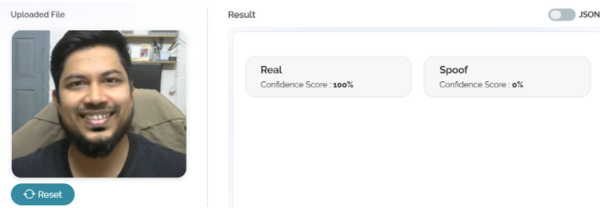


Figure 9: Fake video liveness score from arya

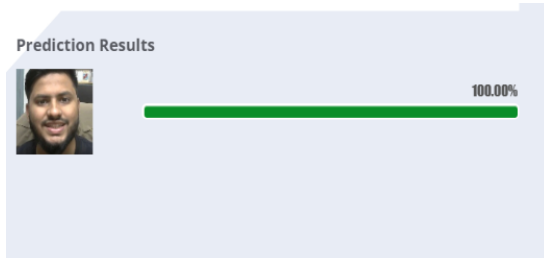


Figure 10: Original video prediction score

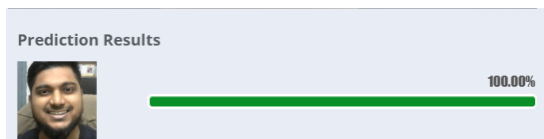


Figure 11: Fake video prediction score



Figure 12: Original video score from deepware

- **Deepware.AI Scores:** We scanned our generated deepfake 13 videos and original 12 videos in Deepware.AI [9]. Detects avatarify deepfake on the provided video clip.

5.3 Effects of Video Resolution

In our study we found that video resolution plays a crucial role in the detection performance of deepfake detectors. We found that higher-resolution videos could bypass detection algorithms more successfully compared to lower-resolution ones.

- **High-Resolution Videos:** These high-resolution deepfakes look more realistic. With their smooth pixel transitions, they

Model Results

Avatarify: SUSPICIOUS(77%)

Deepware: SUSPICIOUS(68%)

Seferbekov: NO DEEFAKE DETECTED(29%)

Ensemble: NO DEEFAKE DETECTED(45%)

Figure 13: Fake video score from deepware

can avoid detection systems that rely on pixel-level inconsistency analysis because of the fine details they retain. We found that resolutions over 720p videos were able to overcome deepfake detectors.

- **Low-Resolution Videos:** Conversely, videos with lower resolution got caught as fraudulent. Visible distortions and artifacts were visible in the lower quality, making it easier for detection algorithms to recognize them as modified video. We tested deepfakes on Arya.ai after reducing the resolution to 240p, which were identified as fraudulents.

6 Discussion

Through this study we found that current deepfake and liveness detection algorithms are unable to detect advanced face manipulations. These systems' vulnerability becomes evident when tested against sophisticated deepfake generation techniques, particularly in the context of eKYC.

6.1 Comparison with Existing Literature:

While prior studies concentrate on media manipulation detection or general biometric systems. Our findings go beyond this by focusing on the relatively unexplored subject of deepfake generation and its effects on eKYC systems vulnerability 2.

6.2 Addressing Research Questions:

The study confirms the vulnerability of certain eKYC systems to deepfake attacks.

- The study shows that advanced deepfake technologies can actually bypass eKYC systems. Several publicly available liveness and deepfake detection techniques were unable to detect the deepfakes produced by the inswapper_128 and buffalo_1 models. This demonstrates how severe weaknesses in eKYC systems can be exploited by modern deepfake techniques.
- Our examination revealed some serious shortcomings in today's deepfake detection systems. Most eKYC do not have liveness and deepfake detection built in them. They only have face recognition that matches their face with national ID. Very few systems have only liveness detection, which can be easily bypassed using any low-quality deepfakes. Though

Table 2: Qualitative Comparison: Our Study vs Existing Literature

Aspect	Our Study	Existing Literature
Research Focus	First to assess deepfake generation's impact on eKYC systems , particularly in Bangladesh.	Studies like <i>eKYC-DF</i> [11] and systematic reviews [17] focus on dataset creation and general detection methods without evaluating real-world eKYC vulnerabilities.
Deepfake Generation Models	Utilizes inswapper_128 and buffalo_1 embeddings for high-fidelity, identity-preserving face swaps.	Papers such as [17] discuss GAN-based face swapping but don't integrate advanced recognition embeddings like buffalo_1 .
Dataset Utilization	Evaluates deepfakes using locally captured, context-specific data for eKYC systems in Bangladesh.	The <i>eKYC-DF</i> dataset [11] provides a large-scale collection of deepfake videos but lacks the specific application context and real-world deployment evaluation.
Realism & Identity Fidelity	Achieves high realism and identity consistency without requiring green screens, leveraging advanced generation techniques.	Previous works [17] highlight the challenges of detecting deepfakes but don't emphasize identity preservation in generation. Also, <i>eKYC-DF</i> dataset [11] used green screen in their process.
Practical Relevance	Provides actionable insights into how advanced deepfake generation can compromise deployed eKYC systems.	While datasets like <i>eKYC-DF</i> [11] and detection studies [1] contribute to understanding deepfake detection, they don't directly address the practical implications on deployed systems.

some can detect potential inconsistencies in deepfakes, with advanced techniques they can be fooled as well. This shows how poorly current algorithms handle high-resolution videos and subtle manipulations.

6.3 Limitations:

While this study offers important contributions, there are certain limitations that need to be acknowledged:

- *Scope of Evaluation:* The results couldn't be applied to all eKYC systems because the investigation was restricted to publicly accessible detection systems. Proprietary and more

sophisticated detection platforms, which are frequently utilized by financial institutions and large enterprises, were not assessed.

- *Dataset Limitations:* A small number of publicly accessible platforms were used to test the deepfakes. A more comprehensive understanding of the vulnerabilities in eKYC systems would result from expanding the dataset to include private datasets and a wider range of detection methods.

7 Conclusion

Our study shows how adversarial machine learning can cause serious damage to critical infrastructure like fintech. As AI is still booming, more realistic deepfakes will emerge in the future. With numerous social media apps, our data is not safe anywhere on the internet. It became so diversified that anyone can become a victim of these impersonating attacks. Additionally with the shortcomings in eKYC, deepfakes cause havoc.

Future studies should concentrate on strengthening the real-time analytical capabilities of eKYC systems. Additionally, more focus should be given to incorporating multi-modal detection techniques and contextual elements to increase the robustness of liveness detection. Lastly, we should be very careful about sharing my personal information on public sites and social media.

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